

Task 1 Report

Human Rule Formalization of §47 MBO using the RASE Methodology

Course: Management von Informatikprojekten -- RASE / Rule Formalization Project

Team Members: Aryana Hami * Othman * Samih

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1. Introduction

1.1 Background

The construction industry relies heavily on legal and regulatory documents. These regulations are typically written for human interpretation, making them difficult to process automatically using computer systems.

To address this challenge, the **RASE methodology** provides a structured approach for converting legal text into machine-readable representations. RASE separates legal requirements into four categories:

Ca te go ry	Meaning	Description
R	Requirement	The measurable rule that must be satisfied
A	Applicability	The entity or condition to which the rule applies
S	Selection	A specific sub-set or option within an applicability
E	Exception	A case or condition excluded from the rule

This structure enables legal requirements to be transformed into formal rules that can later be evaluated by software systems.

1.2 Project Objective

The objective of this project is to manually analyse **§47 of the Musterbauordnung (MBO)** using the RASE methodology and produce a correct human-generated reference model.

This reference model will later be used in **Task 2** to evaluate whether Large Language Models (LLMs) such as ChatGPT, Claude, Gemini, and Copilot can reproduce the same annotation automatically.

2. Description of the Selected Clause

2.1 Original Regulation -- §47 Abs. 1 MBO

The following is the original German text of §47 Abs. 1 Musterbauordnung (MBO):

(Satz 1)	Aufenthaltsräume müssen eine lichte Raumhöhe von mindestens 2,40 m haben.
(Satz 2)	Aufenthaltsräume im Dachraum müssen eine lichte Raumhöhe von mindestens 2,20 m über mindestens der Hälfte ihrer Netto-Raumfläche haben; Raumteile mit einer lichten Raumhöhe bis zu 1,50 m bleiben außer Betracht.
(Satz 3)	Die Sätze 1 und 2 gelten nicht für Aufenthaltsräume in Wohngebäuden der Gebäudeklassen 1 und 2.

2.2 Main Topics Covered by §47

This clause covers the following main topics:

- * **Habitable rooms (Aufenthaltsräume):** The general subject of the clause
- * **Ceiling heights:** Minimum clear room height of 2.40 m for standard rooms
- * **Attic rooms (Dachraum):** Special reduced height rule of 2.20 m over at least 50% of the floor area
- * **Building class exceptions:** Rules in Satz 1 and 2 do not apply to residential buildings of classes 1 and 2

2.3 Explanation of the Clause

Satz 1 -- General Rule: All habitable rooms (Aufenthaltsräume) must have a minimum clear room height (lichte Raumhoehe) of at least 2.40 m. This is the baseline requirement that applies to every habitable room in any building.

Satz 2 -- Special Rule for Attic Rooms: Habitable rooms located in the roof space (Dachraum) are subject to a reduced minimum height of 2.20 m, but this height must be achieved over at least half (50%) of the net floor area. This sentence functions as an ExceptionSection -- it overrides Satz 1 for the specific case of attic rooms. The additional statement that room parts with a height below 1.50 m are disregarded defines the method for measuring the net floor area; it is not a legal requirement in itself.

Satz 3 -- Building Class Exception: The requirements of Satz 1 and Satz 2 do not apply to habitable rooms in residential buildings (Wohngebäude) of building classes 1 and 2 (Gebäudeklassen 1 und 2). This sentence is represented in ACCORD as an ExceptionSection. Within this ExceptionSection, the terms "Wohngebäuden" and "Gebäudeklassen 1 und 2" are annotated as Requirement tags that define the scope of the exception.

3. Manual RASE Analysis

3.1 Sentence 1

"Aufenthaltsräume müssen eine lichte Raumhöhe von mindestens 2,40 m haben."

Analysis:

- * **Applicability (A1):** Aufenthaltsräume -- the rule applies to all habitable rooms
- * **Requirement (R1):** lichte Raumhoehe $\geq$ 2.40 m

3.2 Sentence 2

"Aufenthaltsräume im Dachraum müssen eine lichte Raumhöhe von mindestens 2,20 m über mindestens der Hälfte ihrer Netto-Raumfläche haben; Raumteile mit einer lichten Raumhöhe bis zu 1,50 m bleiben außer Betracht."

Analysis:

- * **Applicability (A2):** Aufenthaltsräume
- * **Applicability (A3)** im Dachraum
- * **Requirement (R2):** lichte Raumhoehe $\geq$ 2.20 m
- * **Requirement (R3):** Over at least 50% of net floor area
- * **Note (NOT a RASE element):** The 1.50 m statement defines a measurement methodology and therefore receives no RASE tag.

3.3 Sentence 3

"Die Sätze 1 und 2 gelten nicht für Aufenthaltsräume in Wohngebäuden der Gebäudeklassen 1 und 2."

Analysis:

- * **ExceptionSection:** The entire sentence forms an ExceptionSection because it excludes a category of buildings from the requirements of Sentence 1 and Sentence 2.
- * **Requirement (R4):** Wohngebäude
- * **Requirement (R5):** Gebäudeklassen 1 und 2

4. Interpretation Challenges

Challenge 1 -- Is Sentence 2 a Requirement or an Exception?

This section documents the key semantic challenges encountered during the RASE annotation process.

Initial interpretation: Sentence 2 initially appeared to be another independent requirement running in parallel with Sentence 1.

Resolved: After reviewing the clause structure and receiving expert feedback from Nick Nisbet, it became clear that Sentence 2 functions as a special rule for attic rooms (Exception Section) and therefore overrides the general rule from Sentence 1, rather than extending it.

Challenge 2 -- The 1.50 m Statement -- "bleiben ausser Betracht"

Initial interpretation: The sentence "Raumteile mit einer lichten Raumhoehe bis zu 1,50 m bleiben ausser Betracht." was initially interpreted as an Exception element and marked up accordingly.

Resolved: After receiving Nick Nisbet's final refinement e-mail (31 May 2026), it became clear that this sentence defines a method of measurement for net floor area -- not a legal requirement on room design. It should therefore NOT be marked up as any RASE element at all.

5. Expert Feedback Integration -- Nick Nisbet (AEC3 / PhD Researcher)

Throughout the project we were in e-mail contact with **Nick Nisbet** (AEC3 UK Ltd, PhD researcher), an expert in RASE methodology and rule formalization. His feedback was essential in refining our final annotation.

Point 1: Aufenthaltsräume is itself an Application

The title "Aufenthaltsräume" at the beginning of the clause must itself be marked as an Applicability element. It cannot be treated as plain introductory text.

Point 2: Sentence 2 should be an ExceptionSection

Sentence 2 (the Dachraum rule) must be treated as a special ExceptionSection that overrides the general rule from Sentence 1. It is not an independent parallel requirement.

Point 3: The 1.50 m statement should NOT be marked up

The sentence about room parts with a clear height up to 1.50 m defines the method of measuring net floor area. It is not a requirement on the design and should receive no RASE mark-up at all. Nick's exact words: "It is part of a definition of the method of measurement, not a requirement on the design."

*"By the way, it is very kind of you to say it, but I am not a professor, I am still writing my PhD!
-- Nick Nisbet, e-mail 29 May 2026"*

6. Final RASE Model

The following table presents the final validated RASE annotation of §47 Abs. 1 MBO, incorporating all expert feedback from Nick Nisbet.

ID	Type	Clause	Content	Metric / Property
0	Applicability	Satz 1	Applicability Title Aufenthaltsräume	<i>type = Aufenthaltsraeume</i>
A1	Applicability	Satz 1	Aufenthaltsräume	<i>type = Aufenthaltsraeume</i>
R1	Requirement	Satz 1	Min. clear height 2.40 m	<i>lichteRaumhoehe >= 2.40 m</i>
A2	Applicability	Satz 2	Aufenthaltsräume	<i>type = Aufenthaltsraeume</i>
A3	Applicability	Satz 2	im Dachraum	<i>Type = im Dachraum</i>

ID	Type	Clause	Content	Metric / Property
R2	Requirement	Satz 2	Min. clear height 2.20 m	<i>lichteRaumhoehe >= 2.20 m</i>
R3	Requirement	Satz 2	Over >= 50% of net floor area	<i>nettoRaumflaeche >= 0.50 ratio</i>
R4	Requirement	Satz 3	Wohngebäuden	<i>type = Wohngebaeuden</i>
R5	Requirement	Satz 4	Gebäudeklassen 1 und 2	<i>type = Gebaeudeklasse <=2</i>

Note: The sentence about room parts with a clear height up to 1.50 m (Satz 2) is intentionally excluded. Per expert feedback, it defines a measurement methodology, not a legal requirement.

7. Machine-Readable Rules

The final RASE model was transformed into IF/THEN logic rules suitable for computational interpretation. These rules serve as the machine-readable representation of §47 Abs. 1 MBO.

Rule R1

```
IF
Room = Aufenthaltsraum
AND Room != Dachraum
AND BuildingClass NOT IN {1, 2}
THEN
Height >= 2.40 m
```

Rule R2

```
IF
Room = Aufenthaltsraum
AND Room = Dachraum
AND BuildingClass NOT IN {1, 2}
THEN
AreaWithHeight >= 2.20 m
AND >= 50% of NetArea
```

8. Final HTML Annotation

This section presents the final RASE-annotated HTML produced using the ACCORD Rule Formalization Tool (arc.salleurl.edu). The annotation was validated with expert feedback from Nick Nisbet.

8.1 ACCORD Tool -- Visual Output

The following screenshot shows the final annotation as displayed in the ACCORD tool. RASE elements are highlighted using the standard color coding:

Color	RASE Type	Meaning
Blue background	Requirement	Measurable rule that must be satisfied
Green underline	Applicability	Entity or condition the rule applies to
Yellow background	Exception Section	Section containing exception to the general rule

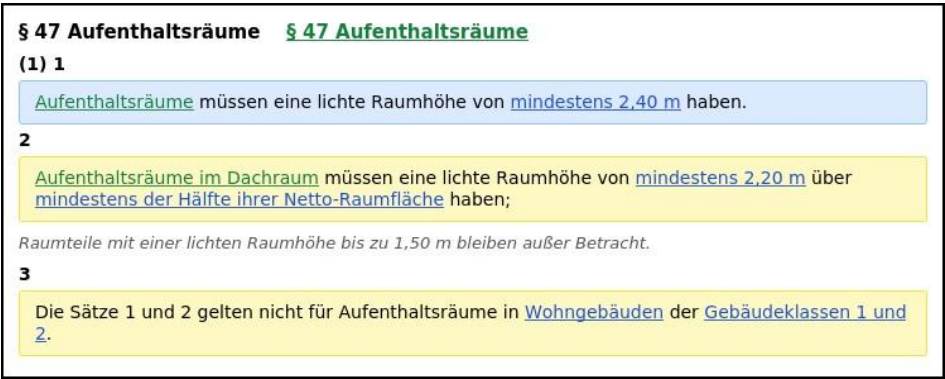


Figure 1: ACCORD Tool output showing §47 with RASE color-coding. Green = Applicability, Blue = Requirement, yellow background = ExceptionSection.

8.2 HTML Mark-up Structure

The ACCORD tool exports the annotation as HTML using data attributes to encode RASE elements. The key HTML structure is as follows:

```
<section data-identifier="3" title="$ 47 Aufenthaltsräume">
  <span data-rasetype="Applicability" data-raseproperty=":type == :Aufenthaltsraeume">
    Aufenthaltsräume
  </span>
  <div data-rasetype="RequirementSection" data-identifier="2.1">
    <span data-rasetype="Requirement" data-raseproperty=":lichteRaumhoehe >= 2.40 :metres">
      mindestens 2,40 m
    </span>
  </div>
```


</section>

The key HTML attributes used are: **data-rasetype** (the RASE classification: Application, Requirement, RequirementSection, ExceptionSection) and **data-raseproperty** (the machine-readable metric or condition).

9. Human vs LLM Interpretation

In addition to the manual RASE annotation, AI-assisted analysis was conducted using Large Language Models (LLMs). The generated outputs were compared against the manually produced RASE model.

#	Aspect	Human Annotation	LLM Tendency
1	1.50 m statement	Correctly excluded -- identified as measurement definition, not a rule	Often marks up as Exception or Requirement element
2	Sentence 2 structure	Correctly treated as ExceptionSection overriding Sentence 1	Often treated as independent parallel Requirement
3	Applicability scope	Aufenthaltsräume marked as Application at clause level	May omit the clause-level Application tag
4	Nested sections	Correctly uses RequirementSection and ExceptionSection nesting	May produce flat structure without nesting

The comparison showed that AI can correctly identify many RASE elements. However, AI systems consistently struggle with semantic distinctions such as measurement definitions versus legal requirements, and recognising override relationships between sentences. This highlights the importance of human validation and motivates the structured evaluation planned for Task 2.

10. Reflection

10.1 Lessons Learned

This project provided practical experience in legal rule formalization and demonstrated the challenges of translating natural language regulations into structured representations.

The most important lesson was that **legal interpretation requires understanding meaning and context**, not just keyword recognition. The 1.50 m measurement statement is a clear example: it reads like an exception but is actually a definitional rule about area calculation.

The collaboration with Nick Nisbet was invaluable -- it confirmed that even experienced annotators need multiple rounds of review to produce a correct RASE model.

10.2 Difficulty Log

Difficulty	Solution
CSS colors missing from HTML	Solved by copying the two CSS link lines from a working example file

Strange div boxes in ACCORD output	Solved by restarting from the original PDF and re-doing the work
Classification of Sentence 2	Solved through expert feedback: it is an ExceptionSection, not a Requirement
1.50 m statement classification	Solved through Nick's final e-mail: it is a measurement definition, not a RASE element
Initially selected §36 (too complex)	Solved by switching to §47 on Nick's advice

11. Preparation for Task 2

The validated RASE model developed in Task 1 serves as the ground-truth reference for Task 2.

In the next phase, different LLMs (e.g. ChatGPT, Claude, Gemini and Copilot) will be evaluated regarding their ability to reproduce the same RASE annotation automatically. The resulting annotations will be compared against the human-generated reference model.

Task 1 -- Summary of Deliverables

RASE Structure:

- 1 clause-level Applicability
- 4 Applicability tags
- 5 Requirement tags
- 1 RequirementSection
- 2 ExceptionSections
- No RASE tag assigned to the 1.50 m measurement statement. HTML mark-up produced via ACCORD Rule Formalization Tool . Validated with expert feedback from Nick Nisbet (3 rounds of revision) Machine-readable IF/THEN rules for Rule R1 and Rule R2

Ground-truth reference model ready for LLM evaluation in Task 2